

Bulk Mailing Tool

A Mail-Scheduling System for GYWS, IIT Kharagpur

Megha Singh

August 2025

Objective

GYWS is a student-run NGO at IIT Kharagpur that runs donor outreach and awareness campaigns over email. Sending these campaigns manually from a single inbox is slow and trips spam filters, since providers flag bursts of near-identical messages from one sender. The Bulk Mailing Tool was built to let a small team schedule and send large recipient lists in a controlled, trackable way, reducing manual effort and spam-filter risk.

What was built

The system is split into two services:

- **Frontend** — a React (Vite) dashboard, styled with Tailwind CSS, for composing campaigns with a rich-text editor, uploading CSV recipient lists, managing sender “agents,” and reviewing send history.
- **Backend** — a FastAPI service backed by MongoDB. It exposes JWT-authenticated endpoints to create sender agents, send an email immediately, or schedule a single email or a full batch for future delivery. Scheduled jobs are handled by APScheduler and dispatched through `smtplib` over SMTP. Each recipient’s message supports placeholder substitution (e.g. `{{name}}`) so a single template can be personalized per recipient, and every send is logged with its delivery status.

Scope note

This write-up accompanies a CV entry that also references a Redis/Bull-based queue for per-user rate limiting and a Mistral-AI-generated content-variation step to reduce duplicate-content spam flags. Those pieces were part of the intended design and were evaluated during development; the codebase in this repository reflects the FastAPI + APScheduler implementation that was actually shipped and used to send GYWS’s campaigns. The rate-limiting and content-variation ideas remain a natural next iteration on top of the current scheduler.

Outcome

The tool was used to send donor and outreach campaigns for GYWS, supporting multiple scheduled batches with per-recipient personalization and delivery tracking, in support of the organization’s fundraising and awareness efforts during 2025.

Repositories

- Consolidated repository: <https://github.com/MeghaSinghall/bulk-mailing-tool>